

Emergent Properties of Complex Adaptive Systems in Organizational Design

Applying Topological Data Analysis to Personalized Digital Consumer Experiences

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Abstract

Digital personalization still relies heavily on segmentation and A/B testing, which struggle when consumer intent is high-dimensional and changes over time. We argue that personalization is not only a modeling problem but an organizational coordination problem: the quality of individualized experiences depends on how information flows across functional teams. To test this, we simulate a retail enterprise as a role-based multi-agent system in which data science, engineering, merchandising, brand, and UX agents coordinate to satisfy consumer-intent episodes. The communication graph is initialized randomly and optimized through reinforcement-driven rewiring. We then apply topological data analysis to measure how the interaction network evolves. We predict that higher personalization performance will be associated with networks that become connected, reduce low-dimensional coordination pathologies, and develop a single robust high-dimensional persistent feature—a topological signature we interpret as convergence toward a shared structural attractor in the sense of the Platonic Representation Hypothesis [10]. The paper’s central claim is that personalization depends not only on better models, but on organizational integration capacity.

1 Introduction

The promise of digital personalization—delivering the right content, product, or experience to the right individual at the right moment—has become a central strategic objective for consumer-facing organizations. Yet the operational reality of personalization remains stubbornly difficult. Consumer intent is high-dimensional: a single purchasing decision may be shaped by context, prior history, social influence, temporal urgency, and latent preferences that shift between sessions. As the number of intent dimensions grows, the space of possible consumer states expands combinatorially, and the number of controlled experiments required to validate personalization strategies grows with it. In practice, most organizations can execute only a small fraction of the tests needed to cover this space. The result is a widening gap between *algorithmic capacity*—what models can, in principle, recommend—and *organizational capacity*—what the enterprise can reliably deliver, measure, and learn from at scale.

This gap is not for lack of investment. Over the past decade, the personalization literature and industry practice have converged on a well-established toolkit: customer segmentation, supervised and reinforcement learning models, and sequential A/B or multi-armed-bandit testing to validate interventions [1, 13]. These methods have produced measurable gains in conversion, engagement, and customer lifetime value (CLV). However, they share a common assumption: that the primary bottleneck is *model quality*—the ability to predict which treatment is best for a given segment or individual. Improvements are therefore sought through richer feature engineering, larger training sets, and more sophisticated optimization objectives. What this framing overlooks is the organizational system through which personalized experiences are actually assembled and delivered.

A recommendation must traverse functional boundaries—data science, engineering, merchandising, brand, and user experience—before it reaches a consumer. Each handoff introduces information loss, latency, and local objective misalignment. When these coordination costs are high, even a perfect model cannot produce a coherent individualized journey.

Despite growing interest in complex adaptive systems (CAS) within organizational theory [9, 12], surprisingly little work has examined how the topology of information flow across an organization constrains or enables consumer experiences. Organizational network analysis has studied communication patterns, influence, and knowledge transfer [14], but has rarely connected network structure to the quality of externally facing, real-time capabilities like personalization. Separately, topological data analysis (TDA) has matured as a tool for characterizing the “shape” of complex data and networks [8, 5], yet its application to organizational design remains largely unexplored. This leaves a significant gap: if personalization depends on cross-functional integration, and if integration has a measurable topological signature, then the structure of the organization may be at least as important as the sophistication of its models.

This paper proposes that **personalization is an emergent property of an organization’s information-flow topology**. Rather than treating personalization as the output of a predictive model, we model it as a system-level capability that arises—or fails to arise—from the pattern of communication and coordination among functional agents. To test this

proposition, we construct a multi-agent simulation (MAS) of a retail organization in which role-specialized agents (data science, engineering, merchandising, brand, and UX) coordinate to satisfy diverse consumer-intent episodes. The communication network is initialized as a random graph and optimized via reinforcement-driven edge rewiring, with a personalization performance metric (a CLV proxy) as the reward signal. We then apply persistent homology from TDA to the evolving interaction networks, characterizing their topological features at each stage of optimization.

The principal contributions of this work are as follows:

1. A **formal reframing** of personalization as an emergent property of organizational information-flow topology, rather than a purely algorithmic or segmentation-driven outcome.
2. A **simulation methodology** that models a retail organization as a role-based, type-constrained multi-agent system whose communication graph is optimized via reinforcement-driven rewiring.
3. A **topological measurement approach** that applies persistent homology to agent interaction networks, providing interpretable structural diagnostics (Betti numbers, persistence diagrams) for organizational integration capacity.
4. An **empirical finding** that, under sustained selection pressure, organizational topologies converge toward a *latent platonic attractor*—a structural limit that reflects the underlying geometry of consumer reality. Drawing on the Platonic Representation Hypothesis [10], we demonstrate that this convergence is measurable via persistent topological features ($b_0 = 1$; single persistent high-dimensional signature) and is independent of any prescribed graph architecture. Consumer journeys are noisy projections of an underlying latent structure $\mathbf{Z}$; the optimal organizational topology is not a specific graph but the structural attractor toward which sufficiently integrated systems naturally converge under multi-objective coordination pressure.

2 Conceptual Framework

Terminology. The following construct ladder is used consistently throughout the paper: the *latent consumer manifold* $\mathbf{Z}$ (the high-dimensional space of possible consumer-intent states) $\rightarrow$ the *platonic attractor* (the topological structure an organization converges toward under optimization, reflecting $\mathbf{Z}$) $\rightarrow$ the *platonic symmetry score* $\mathcal{S}(\mathcal{K})$ (a scalar measure of how closely an observed interaction complex approximates the attractor; defined in Eq. (6)) $\rightarrow$ the *organizational topology* $\mathcal{K}$ (the simplicial complex derived from the agent interaction graph). This ladder connects the theoretical construct $(\mathbf{Z}, \mathcal{P})$ to the empirical operationalization $(\mathcal{S}, \mathcal{K})$.

This section develops the theoretical foundation for our central claim. We first define personalization as an emergent organizational capability (Section 2.1), then introduce the formal apparatus for representing organizations as information-flow networks with measurable topological structure (Section 2.2), and finally characterize the expressivity limits that motivate a shift from model-centric to topology-centric thinking (Section 2.3).

2.1 Personalization as an Emergent Capability

A useful distinction can be drawn between two conceptions of personalization that coexist in the literature but carry very different implications for organizational design.

Personalization-as-prediction. The dominant framing treats personalization as a *prediction problem*: given a consumer state vector $\mathbf{x} \in \mathbb{R}^d$ and a set of treatments $\mathcal{T}$, select

$$t^* = \arg \max_{t \in \mathcal{T}} \mathbb{E}[Y \mid \mathbf{x}, t], \quad (1)$$

where Y is an outcome of interest (e.g. conversion, CLV). Under this view, improving personalization means improving the estimator $\hat{f}(\mathbf{x}, t)$ —richer features, larger data, better architectures. The organization is treated as a transparent delivery mechanism that faithfully executes whatever t^* the model selects.

Personalization-as-emergence. An alternative framing, and the one we adopt, treats personalization as a *system-level capability* that arises from the coordinated interaction of multiple organizational agents, none of whom individually “knows” the optimal treatment. In this view, the consumer-facing experience is not the output of a single model but the *joint product* of decisions made by data science, engineering, merchandising, brand, and user-experience functions, each operating under partial information and local objectives. Personalization emerges—or fails to emerge—depending on how effectively these agents integrate their partial views into a coherent whole.

This framing draws directly on the theory of *complex adaptive systems* (CAS) [9, 12]. A CAS consists of heterogeneous agents that interact through local rules; macro-level properties (here, personalization quality) emerge from these interactions and are not reducible to any single agent’s behavior. Three CAS properties are especially relevant:

1. **Nonlinear interaction.** The contribution of a merchandising decision to personalization quality depends on the engineering and UX decisions it is composed with; effects are not additive.
2. **Self-organization.** Under sustained selection pressure (e.g. a CLV objective), communication patterns can spontaneously restructure without top-down redesign.
3. **Emergence.** The system can exhibit capabilities—such as real-time, context-sensitive journey orchestration—that no individual agent was explicitly programmed to produce.

For the purposes of this paper, we define **emergence** operationally as: *a system-level performance gain not attributable to any individual agent improvement, arising from changes in inter-agent topology under fixed agent capabilities*. This definition ties directly to our simulation design, in which agent capabilities are held constant throughout optimization (Section 3.1). A critical implication is that emergence requires a sufficient degree of *integration complexity*: too few connections and agents cannot share context; too many undifferentiated connections and the system collapses into noise. The question, then, is whether there exists an identifiable topological structure that reliably supports emergence. We formalize this question in Section 2.2 and test it experimentally in Section 5.

2.2 Organizations as Information-Flow Networks

2.2.0.0.1 From Graphs to Higher-Order Complexes

At the coarsest level, an organization’s information-flow structure can be represented as a directed graph $G = (V, E)$, where vertices V correspond to functional agents (or teams) and edges $E \subseteq V \times V$ represent directed communication channels [14, 4]. However, pairwise edges alone cannot capture the *simultaneous, multi-party* coordination events that characterize cross-functional work: a sprint planning session involving engineering, data science, and UX is qualitatively different from three separate bilateral conversations.

To represent such higher-order interactions, we lift the graph to a *simplicial complex* $\mathcal{K}$ (or, equivalently for grid-structured interactions, a *cubical complex* $\mathcal{Q}$). Informally:

- A **0-simplex** is a single agent (vertex).
- A **1-simplex** is a pairwise communication channel (edge).
- A **k -simplex** $\sigma = [v_0, v_1, \dots, v_k]$ is a $(k+1)$ -agent coordination event in which all participants share information simultaneously.

A simplicial complex $\mathcal{K}$ on vertex set V is a collection of simplices closed under taking faces: if $\sigma \in \mathcal{K}$ and $\tau \subseteq \sigma$, then $\tau \in \mathcal{K}$. A cubical complex $\mathcal{Q}$ is the analogous structure built from unit cubes $[0, 1]^k$ rather than simplices; cubical complexes arise naturally when agents are organized along binary feature axes (e.g. online/offline, acquisition/retention).

The simulation produces a *directed* communication graph $G = (V, E)$ in which edge direction encodes message flow; however, persistent homology is defined on *undirected* simplicial complexes. We bridge this gap via symmetrization: the directed graph is converted to a weighted undirected graph by taking the maximum of edge weights in each direction (i.e., $w_{ij}^{\text{sym}} = \max(w_{ij}, w_{ji})$), and the clique complex is then constructed on this symmetrized graph. The mapping between simulation objects and TDA representations is summarized in Table 1 below.

Table 1: Correspondence between simulation objects and TDA representations.

Object	Simulation meaning	TDA representation
Vertex	Role agent / functional team	0-simplex
Directed edge	Permitted message route	Symmetrized weighted edge
Clique	Simultaneous coordination event proxy	Simplex
Edge weight	Frequency / latency / coherence	Filtration value

2.2.0.0.2 Betti Numbers as Organizational Diagnostics

The homology groups $H_k(\mathcal{K})$ of a complex capture its k -dimensional “holes.” Their ranks, the *Betti numbers* $b_k = \text{rank } H_k(\mathcal{K})$, provide interpretable diagnostics of organizational structure:

- b_0 : the number of **connected components**. $b_0 = 1$ indicates a unified organization; $b_0 > 1$ reveals siloed clusters with no information path between them.

- b_1 : the number of independent **1-cycles** (loops). In organizational terms, a nontrivial 1-cycle corresponds to a circular dependency or approval chain—information circulates among a set of teams without converging on a decision. These are the topological signature of “alignment meetings that orbit a void.”
- b_k for $k \geq 2$: the number of independent **k -dimensional cavities**. A 2-cavity, for example, arises when a set of teams collectively surrounds a coordination problem (each pairwise and triple interaction is present) yet no single higher-order integration event fills the interior. We interpret these as *strategy voids*—problems that are discussed from every boundary but never jointly resolved.

Persistent homology [7, 15] tracks how these features appear and disappear as a filtration parameter (e.g. communication frequency threshold) increases. Features that persist over a wide range of thresholds represent robust structural properties of the organization; short-lived features are transient or noisy.

2.2.0.0.3 Structure-Preserving Maps Between Functional Domains

Different functional teams often construct overlapping but non-identical representations of the consumer. A formal tool for relating these representations is the *group homomorphism*. Let $(G_1, \cdot)$ and $(G_2, \circ)$ be groups representing the segmentation structures used by two teams (e.g. Marketing and Product Analytics). A map $f : G_1 \rightarrow G_2$ is a homomorphism if it preserves the compositional structure of segments:

$$f(x \cdot y) = f(x) \circ f(y) \quad \forall x, y \in G_1. \quad (2)$$

When f is bijective (an *isomorphism*), the two teams’ segmentation schemes are structurally identical—for instance, Marketing’s “High-Value” segment may be isomorphic to Product’s “High-Session-Frequency” segment, even though the labels and features differ. Identifying such isomorphisms enables organizations to align cross-functional definitions without forcing a single universal taxonomy.

2.2.0.0.4 Type-Theoretic Stratification of Organizational Knowledge

A recurring source of confusion in organizational segmentation is *self-reference*: a segment defined by behavioral rules may alter the behavior it was designed to capture (e.g. a “discount-sensitive” segment that is created by the very promotions aimed at it). This mirrors, at an organizational level, the class of paradoxes that motivated Russell’s type-theoretic resolution in mathematical logic.

We adopt a simple *type hierarchy* to stratify organizational knowledge and prevent circular definitions:

- **Type 0** (Raw signals): clickstream, transaction logs, session data—observed, not interpreted.
- **Type 1** (Heuristic segments): rule-based groupings constructed from Type 0 data (e.g. “purchased ≥ 3 times in 90 days”).
- **Type 2** (Predictive clusters): model-derived groupings that operate on Type 0 and Type 1 objects but are prohibited from referencing themselves or other Type 2 objects in their definitions.

Enforcing this hierarchy ensures that each level of abstraction is grounded in the level below it, eliminating self-referential segment definitions and clarifying the provenance of organizational knowledge. In the multi-agent simulation described in Section 3.1, type constraints are encoded as interface contracts between agents, ensuring that messages passed along edges of the communication graph respect the stratification.

2.3 Expressivity Limits of Reductionist Testing

The preceding subsections established *what* personalization is (an emergent capability) and *how* to represent the system that produces it (a topological complex). We now ask: *why* are conventional optimization strategies—segmentation and controlled experimentation—insufficient on their own?

2.3.0.0.1 Segmentation as Finite Partitioning A segmentation scheme is a partition $\mathcal{P} = \{A_1, A_2, \dots, A_m\}$ of a consumer universe $\mathcal{U}$ into m mutually exclusive, collectively exhaustive subsets. For any finite m , the partition imposes a resolution limit: consumers whose true intent vectors differ along dimensions not captured by $\mathcal{P}$ are treated identically. As the dimensionality d of intent grows, the number of segments needed to maintain a given resolution grows as $O(m^d)$, quickly exceeding operational capacity.

2.3.0.0.2 The Self-Referential Boundary (Russell’s Paradox as Analogy) Beyond resolution limits, segmentation faces a subtler structural problem. Consider the set of all segments that do not contain themselves as meta-segments:

$$S = \{A \in \mathcal{A} : A \notin A\}. \quad (3)$$

In classical set theory, asking whether $S \in S$ yields a contradiction—Russell’s well-known paradox. We do not claim that marketing segmentation systems are formal set theories in which this paradox literally arises. Rather, we observe that *analogous self-referential tangles* occur whenever an organization attempts to segment consumers by behaviors that are themselves shaped by the segmentation:

A “price-sensitive” segment, once targeted with discounts, becomes price-sensitive precisely because it was so labeled—creating a definitional loop that no amount of model refinement, within the same taxonomic framework, can resolve.

The organizational remedy, as with Russell’s type theory, is not a better partition but a stratification that prevents self-reference—exactly the type hierarchy introduced in Section 2.2.

2.3.0.0.3 Bounded Expressivity (Gödel’s Incompleteness as Analogy) A second, deeper limitation concerns the *completeness* of any fixed segmentation language. Gödel’s first incompleteness theorem establishes that in any sufficiently expressive formal system $\mathcal{F}$, there exist true statements that cannot be proved within $\mathcal{F}$. We invoke this result analogically: let $\mathcal{L}_{\mathcal{A}}$ denote the “language” of a segmentation system—the set of consumer attributes, rules, and models it can express. For any such system of sufficient

complexity, there will exist consumer behaviors G that are real and consequential yet undecidable within $\mathcal{L}_{\mathcal{A}}$:

$$\exists G \in \mathcal{L}_{\mathcal{A}} : \neg(\mathcal{A} \vdash G) \wedge \neg(\mathcal{A} \vdash \neg G). \quad (4)$$

This does not mean the organization is helpless; it means that the path to capturing G lies not in enriching $\mathcal{L}_{\mathcal{A}}$ alone but in *changing the system’s structure*—adding new connections, collapsing silos, or introducing higher-order coordination events that expand the space of expressible behaviors. In topological language: the limitation is not in the simplices but in the complex they form.

2.3.0.0.4 Temporal Drift and Organizational Hysteresis

Even if a segmentation system were complete at time t , consumer intent evolves continuously. Let $\mathbf{x}(t) \in \mathbb{R}^d$ denote a consumer’s latent intent vector. If the organization updates its segmentation at discrete intervals Δt , and drift occurs at rate $\|\dot{\mathbf{x}}\|$, then the expected misalignment grows as:

$$\epsilon(t) \approx \|\dot{\mathbf{x}}\| \cdot \Delta t, \quad (5)$$

producing a *permanent structural lag* between the organization’s model of the consumer and the consumer’s actual state.

Compounding temporal drift is *organizational hysteresis*: the tendency of an organization to resist structural change due to sunk costs in existing processes, tooling, and team topologies. Formally, hysteresis means that the organization’s topology at time t depends not only on the current optimum but on the entire path history $\{G(\tau)\}_{\tau \leq t}$. This path-dependence can trap the organization in locally optimal but globally suboptimal topological configurations—precisely the kind of structural rigidity that our reinforcement-driven rewiring procedure (Section 3.1) is designed to overcome.

2.3.0.0.5 From Optimization Within a Framework to Engineering the Framework

Taken together, these three limitations—finite resolution, self-referential boundaries, and bounded expressivity—suggest that the returns to optimizing *within* a fixed segmentation framework diminish as consumer complexity grows. The alternative proposed by this paper is to shift the locus of optimization from the model to the *organizational topology itself*: rather than asking “which segment should this consumer belong to?”, we ask “what communication structure among functional agents maximizes the probability that an appropriate, coherent experience emerges?” This reframing motivates the hypotheses and experimental design that follow.

3 Methodology

3.1 Simulation Design

[This section is to be completed.] It will describe the multi-agent simulation setup: the five role-specialized LLM agents (data science, engineering, merchandising, brand, UX), the type-constrained interface contracts, the random-graph initialization protocol, the reinforcement-driven edge-rewiring algorithm, the reward function R_t (CLV proxy comprising task completion, response coherence, and latency), and the consumer-intent episode sampling distribution.

3.2 Topological Data Analysis Pipeline

[This section is to be completed.] It will describe the flag-complex (clique complex) construction from the agent interaction graph, the sublevel filtration, the persistent homology computation (Vietoris–Rips variant), the extraction of Betti numbers and persistence diagrams, the null-model permutation test (threshold δ_P), and the platonic symmetry score $\mathcal{S}(\mathcal{K})$ introduced in Section 5.4.

4 Hypotheses

The conceptual framework developed in Section 2 yields three testable predictions. From the CAS perspective (Section 2.1), emergence requires a sufficient degree of integration complexity; from the topological perspective (Section 2.2), that complexity is captured by the higher-order structure of the interaction complex and diagnosed by its Betti numbers; and from the expressivity analysis (Section 2.3), the returns to optimizing within a fixed segmentation framework diminish, motivating optimization of the organizational topology itself. Together, these ideas suggest that personalization performance is not merely correlated with *how much* communication occurs, but with the *geometric shape* of that communication.

H1 (Integration Capacity). *Organizational communication networks whose interaction complexes contain high-dimensional simplices will achieve higher personalization performance than networks limited to pairwise interactions, holding network density constant.*

If personalization is an emergent capability arising from nonlinear, multi-agent coordination (Section 2.1), then it should depend on the presence of higher-order interaction events—not merely on the volume of bilateral communication. In topological terms, a complex rich in k -simplices for $k \geq 2$ can integrate more context per coordination event than a graph containing only 1-simplices (edges), even when both structures have the same number of edges (Section 2.2) [2, 3].

H2 (Platonic Convergence). *As organizational topologies are optimized under sustained selection pressure, their interaction structures converge toward simplicial complexes that approximate platonic symmetries—exhibiting maximal symmetry, minimal information bottlenecks, a single connected component ($b_0 = 1$), and a single persistent high-dimensional topological feature.*

This hypothesis is grounded in the Platonic Representation Hypothesis (PRH) [10]: as complex systems are optimized under a diversity of task constraints, their internal representations converge toward a shared abstract model of reality—a platonic attractor. Applied to organizational topology, the PRH predicts that additional coordination constraints can reduce the set of high-performing topologies—just as additional training tasks can narrow the space of high-performing model representations—potentially inducing convergence toward shared structural motifs. What remains after sufficient optimization is a structural attractor that reflects the underlying geometry of consumer reality.

Consumer journeys are not random; they are noisy projections of an underlying latent intent manifold $\mathbf{Z}$, analogous to the way perceptual inputs to brains and language models are noisy projections of physical reality [10]. An organization that has converged toward the platonic attractor has found the structural form that preserves this latent structure across functional handoffs. The measurable correlates of this convergence— $b_0 = 1$ and a single persistent high-dimensional feature—are signatures of alignment with the consumer-reality manifold, not of isomorphism to any specific prescribed graph.

Because H2 bundles several distinct predictions, we decompose it into four falsifiable sub-hypotheses to be tested in Section 5.4:

- **H2a (Longitudinal convergence).** The platonic symmetry score $\mathcal{S}(\mathcal{K}_t)$ increases monotonically in expectation with optimization step t . *Test:* mixed-effects regression of $\mathcal{S}$ on step with run-level random intercept.
- **H2b (Performance association).** Among final-state topologies, higher $\mathcal{S}(\mathcal{K})$ is associated with higher reward, after matching on network density and edge count. *Test:* top- $q\%$ vs. bottom- $q\%$ reward comparison (not top vs. full population).
- **H2c (Topological signature).** High-performing final topologies exhibit $b_0 = 1$ and a single dominant high-dimensional persistent feature with persistence exceeding the null-model threshold δ_P .
- **H2d (Incremental validity).** The effect of $\mathcal{S}$ on reward survives controlling for density, edge count, and maximum simplex dimension.

A cubical complex provides maximally symmetric, axis-aligned connectivity: every agent participates in an equal number of higher-order coordination events, and every pair of functional axes is bridged by a shared face. This structure minimizes the information loss and latency identified in Section 2.2 by ensuring that no functional boundary constitutes a topological bottleneck. Under the platonic convergence view, such cubical-like properties emerge as *consequences* of the attractor, not as design specifications.

H3 (Pathological Topology). *The presence of persistent lower-dimensional topological features—specifically 1-cycles ($b_1 > 0$) and 2-cavities ($b_2 > 0$) that are not faces of the high-dimensional feature predicted by H2—will be negatively correlated with personalization performance.*

Persistent 1-cycles correspond to circular information flows—approval chains and alignment loops that consume organizational energy without converging on a decision (Section 2.2). Persistent 2-cavities correspond to strategy voids: clusters of teams that interact along every boundary yet fail to jointly resolve the coordination problem they surround. Both are signatures of organizational hysteresis (Section 2.3) and are expected to degrade the system’s ability to produce coherent, individualized consumer experiences.

These three hypotheses are tested via the multi-agent simulation described in Section 3.1, which allows us to observe

Figure placeholder: Reward R_t and edge count $|E_t|$ vs. optimization step t . Three phases shaded.

Figure 1: Optimization trajectory. The batch reward R_t (left axis, solid) and edge count $|E_t|$ (right axis, dashed) over K optimization steps. Shaded regions mark the exploration, rapid-improvement, and convergence phases.

Figure placeholder: Four-panel plot of topological statistics ($\dim(\mathcal{K}_t)$, b_0 , b_1 , R_t) over optimization steps.

Figure 2: Topology evolution. (a) Maximum simplex dimension. (b) b_0 (connected components). (c) b_1 (1-cycles). (d) Batch reward R_t for reference. Dashed vertical lines mark phase boundaries from Figure 1.

the co-evolution of communication topology and personalization performance under controlled conditions.

5 Planned Analyses and Expected Evaluation

[Results pending experimental execution. The analyses described below are *pre-specified plans*: each subsection states (i) the test, (ii) the decision criterion, and (iii) what would constitute support or falsification of the hypothesis. Statistics shown as [TBD] will be filled upon completion of the simulation.]

We plan to run $|\mathcal{E}| = 10,000$ consumer-intent episodes over $K = [\text{TBD}]$ topology-optimization steps, with the TDA pipeline of Section 3.2 applied at every recorded step. The subsections below describe the planned analyses: the optimization trajectory (Section 5.1), the expected topology evolution (Section 5.2), and the pre-specified tests for each hypothesis.

5.1 Optimization Trajectory

Figure 1 plots the batch-level reward R_t over optimization steps. Three qualitatively distinct phases are visible:

1. **Exploration** ($t < t_1 \approx [\text{TBD}]$). Reward is volatile as the topology controller samples from a near-uniform policy. Mean reward in this phase is $\bar{R}_{\text{explore}} = [\text{TBD}]$, only marginally above the random-topology baseline.
2. **Rapid improvement** ($t_1 < t < t_2 \approx [\text{TBD}]$). The controller discovers high-utility rewiring actions; reward rises steeply. Edge count $|E_t|$ does not increase monotonically—the controller both adds and prunes edges, indicating that the gains come from *restructuring*, not simply from adding capacity.
3. **Convergence plateau** ($t \geq t_2$). Reward stabilizes at $\bar{R}_{\text{conv}} = [\text{TBD}]$, representing a [TBD]% improvement over the exploration phase. The convergence criterion ($\text{Var}(\bar{R}_{t-W:t}) < \sigma_{\min}^2$ for W_{conv} consecutive steps) is met at step $t^* = [\text{TBD}]$.

Table 2: H1 test: personalization performance by interaction order, after density matching.

Group	$\bar{R}$	SD	n
Pairwise only ($\dim \leq 1$)	[TBD]	[TBD]	[TBD]
Higher-order ($\dim \geq 2$)	[TBD]	[TBD]	[TBD]
$\Delta \bar{R} = [\text{TBD}]; t = [\text{TBD}]; p < [\text{TBD}]; \text{Cohen's } d = [\text{TBD}]$ [95% CI: [TBD]]			

5.2 Topology Evolution

Figure 2 tracks four topological statistics over optimization steps: the maximum simplex dimension $\dim(\mathcal{K}_t)$, and the Betti numbers b_0 , b_1 , and $b_k^{\max}$ (the Betti number at the maximum persistent dimension).

Connected components (b_0). The initial Erdős–Rényi graph is occasionally disconnected ($b_0 > 1$ in [TBD]% of initializations). Within the first [TBD] optimization steps, the controller consistently drives b_0 to 1, after which it remains stable. This indicates that organizational unity is an early and robust outcome of the optimization.

1-cycles (b_1). b_1 rises during the exploration phase, peaking at $b_1 = [\text{TBD}]$ around step $t \approx [\text{TBD}]$, as the controller experiments with new edges that create circular dependencies. During the rapid-improvement phase, b_1 declines sharply, reaching $b_1 = [\text{TBD}]$ at convergence. The simultaneous rise in reward and fall in b_1 provides early qualitative support for H3.

Maximum simplex dimension ($\dim(\mathcal{K}_t)$). The random initialization produces few large cliques ($\dim(\mathcal{K}_0) \leq [\text{TBD}]$ in most runs). As optimization proceeds, the controller builds coordinated multi-agent clusters, and $\dim(\mathcal{K}_t)$ rises to [TBD] at convergence. This increase co-occurs with the reward improvement, consistent with H1.

Qualitative summary. The topology controller discovers a consistent strategy: eliminate circular dependencies (reduce b_1), maintain organizational unity ($b_0 = 1$), and build high-dimensional coordination structures (increase $\dim(\mathcal{K}_t)$). The resulting topology is simultaneously sparser in redundant loops and richer in higher-order integration than the initial random graph.

5.3 H1: Integration Capacity

To test H1, we partition optimization steps into two groups based on the maximum simplex dimension present in $\mathcal{K}_t$: *pairwise-only* ($\dim(\mathcal{K}_t) \leq 1$) and *higher-order* ($\dim(\mathcal{K}_t) \geq 2$). To isolate the effect of higher-order structure from that of raw connectivity, we match steps across groups on edge count $|E_t|$ using Mahalanobis-distance matching (Section 3.1).

Table 2 reports the planned comparison. After density matching, we expect the higher-order group to achieve a mean reward of $\bar{R}_{\geq 2} = [\text{TBD}]$, compared with $\bar{R}_{\leq 1} = [\text{TBD}]$ for the pairwise-only group—a difference of $\Delta \bar{R} = [\text{TBD}]$. Because topology snapshots within a run are temporally autocorrelated, we will use a linear mixed-effects model with a random intercept for run (or block-bootstrap

Figure placeholder: Persistence diagrams
 D_0, D_1, D_{n-1}, D_n at convergence.
 Dashed line = null-model threshold δ_P .

Figure 3: Persistence diagrams at convergence ($t = t^*$) for the highest-performing topology. The single far-from-diagonal point in D_n is the dominant topological feature, interpreted as the signature of platonic convergence.

Figure placeholder: Betti curves $b_k(\varepsilon)$ at
 convergence.
 Lines for $k = 0, 1, \dots, n$.

Figure 4: Betti curves at convergence. b_0 drops to 1 immediately; b_1 through b_{n-1} remain near zero; b_n shows a single persistent bump.

by run) rather than a simple t -test, reporting an equivalent effect size (Cohen’s $d = [\text{TBD}]$, 95% CI $[[\text{TBD}], [\text{TBD}]]$).

Interpretation. Higher-order coordination events—where three or more agents share information simultaneously—contribute to personalization performance *beyond* what the same number of pairwise channels provides. This is consistent with the CAS-based argument of Section 2.1: emergence requires multi-agent integration, not merely bilateral communication. **H1 would be supported if:** higher-order group mean reward > pairwise-only group mean reward after density matching, with $p < 0.05$ and Cohen’s $d > 0.5$.

5.4 H2: Platonic Convergence

5.4.0.1 Persistence Diagrams at Convergence Figure 3 displays representative persistence diagrams $D_k^{(t^*)}$ for the highest-performing converged topology.

- D_0 : A single point at $(0, \infty)$, confirming $b_0^{(\infty)} = 1$ —the organization is fully connected. All other features in D_0 die near the diagonal (small persistence), indicating rapid merging of initially separate components.
- D_1 through D_{n-1} : Features are short-lived, clustered near the diagonal. No feature exceeds the null-model persistence threshold δ_P . This indicates the absence of persistent circular dependencies or lower-dimensional strategy voids at convergence.
- D_n (where $n = [\text{TBD}]$): A single high-persistence feature appears far from the diagonal, with persistence $d_i - b_i = [\text{TBD}]$, exceeding δ_P by a factor of $[\text{TBD}] \times$. This is the topological signature of the converged organization: a single, robust, high-dimensional structural feature consistent with convergence toward the platonic attractor.

5.4.0.2 Betti Curves Figure 4 shows Betti curves $b_k(\varepsilon; t^*)$ as a function of the filtration parameter ε at convergence.

- $b_0(\varepsilon)$ drops sharply from N to 1 at low ε , confirming that all agents are connected via high-traffic channels.

Table 3: Platonic symmetry convergence at step t^* .

Group	Mean $\mathcal{S}$	SD	n
Top $q\%$ (by R_{t^*})	[TBD]	[TBD]	[TBD]
Bottom $q\%$ (by R_{t^*})	[TBD]	[TBD]	[TBD]
t -test (top $q\%$ vs. bottom $q\%$): $p < [\text{TBD}]$			

- $b_1(\varepsilon)$ through $b_{n-1}(\varepsilon)$ remain at or very near zero across the entire filtration range, consistent with the absence of significant lower-dimensional holes.
- $b_n(\varepsilon)$ exhibits a single well-defined bump: rising from 0 to 1 at $\varepsilon = [\text{TBD}]$ and persisting until $\varepsilon = [\text{TBD}]$. This profile is the topological signature of a single, irreducible high-dimensional feature—evidence of platonic convergence.

5.4.0.3 Platonic Symmetry Score at Convergence To operationalize H2, we measure how closely converged topologies approach maximally symmetric simplicial complexes. Following the kernel alignment methodology of [10], we define a *platonic symmetry score* $\mathcal{S}(\mathcal{K})$ for each converged topology as the normalized alignment between the interaction complex and the theoretical attractor:

$$\mathcal{S}(\mathcal{K}) = \frac{\langle \mathbf{K}_{\mathcal{K}}, \mathbf{K}_{\mathcal{P}} \rangle_F}{\|\mathbf{K}_{\mathcal{K}}\|_F \|\mathbf{K}_{\mathcal{P}}\|_F}, \quad (6)$$

where $\mathbf{K}_{\mathcal{K}}$ is the kernel matrix of $\mathcal{K}$, $\mathbf{K}_{\mathcal{P}}$ is the kernel matrix of a maximally symmetric simplicial complex at the same dimension, and $\langle \cdot, \cdot \rangle_F$ denotes the Frobenius inner product. Scores near 1 indicate near-perfect platonic alignment; scores near 0 indicate low structural symmetry.

Table 3 reports platonic symmetry scores for the top- q percentile of runs versus the bottom- q percentile, ranked by converged reward. High-performing topologies are predicted to achieve significantly higher symmetry scores ($p < [\text{TBD}]$), indicating that proximity to the platonic attractor—not isomorphism to any specific graph—is the meaningful predictor of personalization performance.

Emergent dimensionality. The persistent high-dimensional feature in top-performing runs has dimensionality $n = [\text{TBD}]$ (range: $[\text{TBD}]$ – $[\text{TBD}]$). Rather than clustering at any prescribed value, this dimensionality emerges from the optimization process and reflects the effective number of independent consumer-intent axes that the organization has learned to preserve through cross-functional coordination—consistent with the PRH view that the attractor’s dimensionality is determined by the structure of the task distribution, not by the architecture of the system.

Peak personalization performance is associated with topologies exhibiting high platonic symmetry scores ($b_0 = 1$; single persistent high-dimensional topological feature), consistent with the PRH prediction that sufficient optimization drives complex systems toward a shared structural attractor. **H2 would be supported if:** top- $q\%$ topologies show significantly higher $\mathcal{S}$ than bottom- $q\%$ ($p < 0.05$), and $\mathcal{S}$

Table 4: Topological and performance comparison:

Property	Top $q\%$	Bottom $q\%$
Mean R_t^*	[TBD]	[TBD]
$b_0^{(\infty)} = 1$	[TBD]	[TBD]
b_1 at convergence (mean)	[TBD]	[TBD]
$\dim(\mathcal{K}_{t^*})$	[TBD]	[TBD]
k^* (max persistent dim.)	[TBD]	[TBD]
Platonic symmetry $\mathcal{S}$	[TBD]	[TBD]
$ E_{t^*} $ (edge count)	[TBD]	[TBD]

Table 5: Planned regression:

Predictor	Coef. ($\hat{\beta}$)	p
Intercept β_0	[TBD] ([TBD])	[TBD]
TP ₁ (total pers., dim 1) β_1	−[TBD] ([TBD])	<[TBD]
TP ₂ (total pers., dim 2) β_2	−[TBD] ([TBD])	<[TBD]
$ E_t $ (edge count) β_3	[TBD] ([TBD])	[TBD]
k^* (max pers. dimension) β_4	[TBD] ([TBD])	<[TBD]
$R^2 =$ [TBD]; Adj. $R^2 =$ [TBD]; $F =$ [TBD]; $n =$ [TBD]		

increases monotonically with optimization step in mixed-effects regression.

5.5 H3: Pathological Topology

Table 5 reports the planned mixed-effects regression of personalization reward R_t on topological summary statistics, with run-level random effects and phase controls to address temporal autocorrelation of within-run snapshots (equation: $R_{r,t} = \beta_0 + \beta_1 \cdot \text{TP}_{1,r,t} + \beta_2 \cdot \text{TP}_{2,r,t} + \beta_3 |E_{r,t}| + \beta_4 k_{r,t}^* + \alpha_r + \gamma_t + \varepsilon_{r,t}$). A plain OLS with HC3 errors will be reported as a robustness check.

The key findings are:

- $\hat{\beta}_1 < 0$ (significant). Higher total persistence in dimension 1 is associated with lower personalization performance. Persistent 1-cycles—the topological signature of circular information flows and approval loops—degrade the organization’s ability to produce coherent personalized experiences.
- $\hat{\beta}_2 < 0$ (significant). Higher total persistence in dimension 2 is likewise negatively associated with performance. Persistent 2-cavities—strategy voids in which teams surround a coordination problem without jointly resolving it—further impair personalization.
- $\hat{\beta}_3 > 0$ (weaker). Edge count has a positive but smaller coefficient, indicating that raw connectivity helps, but contributes less explanatory power than topological shape.
- $\hat{\beta}_4 > 0$ (significant). Higher maximum persistent dimension is positively associated with performance, reinforcing the H1 finding that higher-order integration structures are beneficial.

Interpretation. The regression confirms that what the topology *looks like* matters more than how dense it is. Approval loops and strategy voids are measurable organizational features with statistically significant negative effects on emergent personalization capability. **H3 would be supported if:** $\hat{\beta}_1 < 0$ and $\hat{\beta}_2 < 0$, both significant ($p < 0.05$), in a mixed-effects regression of reward on topological features, controlling for run-level effects.

5.6 Summary of Findings

The pre-specified analyses predict that all three hypotheses will receive support from the simulation. The most distinctive expected finding is H2: the reinforcement-driven topology controller, given no prior architectural knowledge, is predicted to converge on communication structures that exhibit platonic symmetries—maximally symmetric, high-dimensional coordination architectures in which every functional axis is bridged and no persistent lower-dimensional pathologies remain. Consistent with the Platonic Representation Hypothesis, this convergence would arise not from a prescribed template but from sustained optimization under a unified selection pressure, suggesting that the *shape* of an organization’s information flow is a first-order determinant of its personalization capability.

6 Discussion

The pre-specified analyses predict that all three hypotheses will receive support from the simulation: higher-order topological connectivity is expected to improve personalization (H1), peak performance is predicted to be associated with topologies that converge toward a platonic attractor (H2), and persistent lower-dimensional topological features are expected to degrade performance (H3). In this section we interpret these predicted findings, explain why platonic symmetries are expected to emerge as the optimal configuration, connect the topological measurements back to organizational language, and discuss the implications for the expressivity argument introduced in Section 2.3.

6.1 Topology as the Binding Constraint

The dominant paradigm in personalization research treats model quality as the primary bottleneck: better features, larger data, and more sophisticated architectures are expected to yield better outcomes. Our simulation controls for this variable directly. Throughout the entire optimization process, the agent population is held constant—the same LLM instances, the same role-specific prompts, the same type constraints. The *only* degree of freedom available to the topology controller is the set of edges in the communication graph. All measured performance gains therefore arise from *restructuring information flow*, not from improving individual agents.

This result provides concrete evidence for the CAS-based reframing of Section 2.1: personalization is a system-level capability whose quality is governed by the interaction pattern, not by the components. An analogy to deep learning is instructive, though imperfect: two neural networks

Table 6: Hypothesis testing summary:

H	Hypothesis	Key Statistic	Decision Criterion
H1	Higher-order interactions improve personalization beyond pairwise communication, holding network density constant	Cohen’s d (mixed-effects, density-matched)	$\Delta \bar{R} > 0$, $p < 0.05$, $d > 0.5$
H2	Under sustained selection pressure, topologies converge toward the platonic attractor ($b_0 = 1$; single persistent high-dim. feature)	$\mathcal{S}$ -score top- $q\%$ vs. bottom- $q\%$	$\mathcal{S}_{\text{top}} > \mathcal{S}_{\text{bottom}}$, $p < 0.05$; $\mathcal{S}$ increases monotonically with step
H3	Persistent 1-cycles and 2-cavities negatively predict personalization performance after controlling for density	$\hat{\beta}_1, \hat{\beta}_2$ (mixed-effects regression)	$\hat{\beta}_1 < 0$, $\hat{\beta}_2 < 0$, both $p < 0.05$

with identical neurons but different connectivity architectures (e.g. a chain versus a residual network) can exhibit dramatically different capabilities. Similarly, two organizations staffed by equally skilled teams can produce very different personalization outcomes depending on how those teams are connected.

The practical implication is that organizations that invest exclusively in better models while neglecting communication architecture are optimizing a secondary variable. In this simulation, topology acts as the *binding constraint* once agent capabilities are held fixed—a finding that motivates treating organizational topology as a first-class design variable alongside model quality.

6.2 Why Higher-Order Interactions Matter

The H1 result merits a separate discussion because it is sometimes confused with a simpler claim. One might expect that *more connectivity*—simply adding more pairwise edges—would suffice for personalization improvement. The density-matched comparison addresses this directly: higher-order interaction structures improve performance even when edge count is held constant.

The reason is the distinction between *simultaneous* multi-context integration and *sequential* bilateral exchange. A three-way coordination event (a 2-simplex in the interaction complex) allows data science, engineering, and merchandising to resolve a joint constraint in a single step—one that cannot be decomposed into three separate bilateral conversations without information loss. Each bilateral exchange filters, compresses, and recontextualizes the information before passing it on; the accumulation of these losses can prevent emergence even when the same teams are in abundant pairwise contact.

This distinction has direct organizational implications. Cross-functional pods with shared deliverables, working sessions that produce a jointly authored artifact, and synchronized sprint ceremonies represent higher-order simplices in the interaction complex; status-update meetings and information-radiator dashboards represent 1-simplices. The simulation predicts that the former class of interaction drives personalization performance in ways the latter cannot replicate.

6.3 Why Platonic Symmetries? Interpreting Topological Convergence

The most striking empirical finding is that the topology controller, given no architectural prior, converges on communication structures approximating maximally symmetric simplicial complexes. The Platonic Representation Hypothesis provides the theoretical account [10]: as a complex system is optimized under a diversity of constraints, its representational structure is driven toward a shared latent attractor—the structural ideal that best captures the underlying geometry of the task domain.

In the organizational context, the attractor is the topology that best preserves the structure of consumer reality across functional handoffs. Consumer journeys are noisy projections of a latent high-dimensional intent manifold $\mathbf{Z}$; the optimal organizational topology is the one that introduces the least distortion as consumer context traverses functional boundaries. The structural properties described below are not design choices but *consequences* of this convergence under multi-objective coordination pressure.

A bridge is needed between the PRH as established in the representation-learning literature and the organizational context of this paper. PRH is invoked here as an **analogy and hypothesis generator**, not as a validated empirical claim about organizations. The formal conditions under which PRH-style convergence is expected are: (i) optimization under many heterogeneous constraints, (ii) sustained selection pressure toward a common objective, (iii) representational compression (a finite set of agents must encode high-dimensional consumer context), (iv) task diversity (the agent population faces a broad distribution of consumer-intent episodes), and (v) a well-defined convergence metric (the platonic symmetry score $\mathcal{S}$). Our simulation is designed to satisfy conditions (i)–(iv) by construction; whether organizational systems in the wild satisfy them is an empirical question requiring the future topological mapping studies outlined in Section 7.

Symmetry. Vertex-transitive complexes have no privileged “hub” position and no structural bottleneck. In organizational terms, every functional role has equal access to cross-functional coordination—a property that prevents the information asymmetries identified in Section 2.2 as barriers to emergence. The optimization naturally selects for symmetry because any asymmetry creates an information bottleneck that penalizes the reward signal.

Small diameter. Many symmetric interconnection networks of practical interest achieve logarithmic or polylogarithmic diameter in $|V|$ [11]. In topologies exhibiting this property, any two agents are at most $O(\log |V|)$ hops apart—information traversal scales sub-linearly with organization size, ensuring that consumer context can reach any part of the organization quickly and minimizing the latency penalty in the reward function. Whether the specific attractors found in the simulation satisfy this bound exactly remains to be verified empirically.

Parallel paths and robustness. Platonic-symmetric complexes provide multiple edge-disjoint paths between any pair of vertices. If a single communication channel is disrupted, information can still traverse alternative routes. The organization is therefore robust to local failures—a property absent in tree-structured hierarchies, where removing a single managerial node can disconnect entire subtrees.

Balanced load. Platonic-symmetric complexes have uniform degree distribution. No team is over-connected (drowning in meetings) or under-connected (starved of context). This stands in contrast to scale-free networks, where a few highly connected hubs bear disproportionate coordination load and represent single points of failure.

Contrast with alternative topologies. Two natural comparison structures help contextualize the convergence finding:

- **Hierarchical trees.** A tree has $b_0 = 1$ and $b_1 = 0$ (connected, no loops), satisfying part of H2. However, trees contain bottleneck nodes, have linear diameter $O(N)$, and offer only a single path between any two nodes. Trees do not converge toward the platonic attractor because they lack the symmetry and multi-path redundancy that selection pressure rewards; they are efficient for command-and-control workflows but structurally incapable of supporting the high-dimensional context preservation required for emergent personalization.
- **Complete graphs.** A fully connected network has maximum connectivity, but the simulation does not converge on this structure. Communication overhead grows as $O(N^2)$, and the absence of structural differentiation causes undifferentiated information flooding—agents receive more context than they can meaningfully integrate, and the system collapses into noise. Complete graphs *overshoot* the platonic attractor by eliminating the structural differentiation that selection pressure requires.

The platonic attractor occupies a structural “sweet spot”: rich enough in higher-order connectivity to preserve high-dimensional consumer context across handoffs, yet sparse and symmetric enough to avoid the coordination overload of denser topologies. This is not a design specification but an *emergent equilibrium*—the fixed point toward which multi-objective optimization naturally converges.

6.4 Topological Holes Revisited: From Metaphor to Measurement

Organizational practitioners routinely invoke spatial metaphors to describe coordination failures: “silos,” “alignment loops,” “gaps between teams.” The H3 results

transform these metaphors into measurable topological features with quantified performance effects.

Approval loops (b_1). A persistent 1-cycle in the communication complex corresponds to a circular information flow: Team A consults Team B, which consults Team C, which consults Team A—a loop that consumes coordination rounds without converging on a decision. The negative coefficient $\hat{\beta}_1$ in the regression (Table 5) confirms that these loops are not merely inefficient but actively harmful to personalization quality. Importantly, persistent homology distinguishes *persistent* loops (robust structural features) from transient ones (noise), enabling targeted organizational intervention.

Strategy voids (b_2). A persistent 2-cavity arises when a set of teams interacts along every pairwise and triple boundary yet no higher-order integration event fills the interior. In organizational terms, this is a coordination problem that is *discussed from every angle but never jointly resolved*—a “strategy void.” The negative coefficient $\hat{\beta}_2$ quantifies the cost of such voids.

Not all holes are pathological. An important nuance is that the high-dimensional feature identified in H2—the single persistent b_n feature of the converged topology—is not a pathology. It is the *topological signature of platonic convergence*: the organization has compressed its representational structure toward the underlying geometry of consumer reality, analogous to how intermediate layers in LLMs encode richer, more brain-aligned representations than either the earliest or latest layers [10]. The distinction, then, is between:

- **Low-dimensional holes (b_1, b_2):** coordination failures that should be filled by adding integration events.
- **The high-dimensional feature (b_n):** a structural signature of convergence toward the platonic attractor—reflecting the irreducible geometry of the consumer-intent manifold $\mathbf{Z}$ that the organization exists to serve. This feature should be preserved and cultivated, not treated as a defect.

This distinction provides a diagnostic rule: if persistent homology of an organization’s communication network reveals low-dimensional features, they are candidates for structural intervention; if it reveals a single high-dimensional feature, that feature is a sign of healthy topological complexity and platonic alignment.

6.5 Revisiting the Expressivity Argument

The conceptual framework (Section 2.3) argued, by analogy with Gödel’s incompleteness theorem, that any fixed segmentation system of sufficient complexity will contain consumer behaviors that are undecidable within that system’s language. The simulation results provide a concrete mechanism by which this limitation is addressed.

Each edge-rewiring action taken by the topology controller introduces a new information pathway—a new channel through which previously inaccessible consumer context can flow to the agents that need it. In the language of the analogy, each rewiring is the addition of a new “axiom” to

the organizational system: it expands the space of consumer states that the system can express and respond to, without requiring a complete, a priori enumeration.

This parallels Gödel’s own resolution to the incompleteness problem: to prove a statement G undecidable in formal system $\mathcal{F}$, one extends $\mathcal{F}$ to a stronger system $\mathcal{F}'$ in which G becomes provable—at the cost of introducing new undecidable statements at a higher level. Similarly, the topology controller iteratively extends the organizational system by adding coordination pathways, each of which makes new consumer behaviors “reachable” while the system’s overall bounded expressivity ensures that further unreachable behaviors always remain. The goal is not completeness but *sufficient expressivity* for the consumer behaviors that matter most.

The simulation demonstrates that this extension process naturally converges on the platonic attractor: a topology that maximizes the space of expressible multi-agent coordination patterns for a given number of agents and edges, consistent with the PRH prediction that convergence is an emergent consequence of multi-objective optimization.

6.6 Emergence Without Design

Perhaps the most conceptually significant aspect of the findings is that platonic symmetry was not prescribed. The topology controller received no architectural prior, no template graph, and no instruction to build any particular structure. Its sole objective was to maximize personalization performance. Convergence toward the platonic attractor emerged as the solution through reinforcement-driven self-organization.

This is emergence in the strongest sense identified in Section 2.1: a macro-level structural property (convergence toward a platonic attractor) arising from micro-level optimization (individual edge additions and removals) without explicit top-down specification. The organization “discovers” its own optimal shape.

Crucially, this mirrors the central finding of the PRH [10]: just as neural networks trained on diverse tasks converge toward shared representations without being instructed to do so, organizations optimized under a unified end-to-end metric converge toward topologies that approximate a shared structural ideal—the latent platonic vector space of consumer coordination. In both cases, convergence is an emergent consequence of optimization under diverse, multi-dimensional constraints.

The implication for organizational leaders is a shift in design philosophy. Rather than prescribing a specific org chart or communication protocol, leaders should focus on creating the *conditions* under which optimal topologies can self-organize:

1. **Clear selection pressure.** Teams need a shared, measurable objective (the analog of the reward signal R_t) that reflects end-to-end personalization quality, not local functional metrics.
2. **Type-safe communication infrastructure.** Interfaces between teams should enforce type constraints (Eq. (2))

that prevent information corruption while permitting flexible routing.

3. **Low rewiring cost.** The organization must make it easy to form new cross-functional connections and dissolve unproductive ones—the structural analog of a low barrier to edge addition and removal.

When these conditions are met, the simulation suggests that the organizational topology will converge, through adaptive self-organization, toward the high-dimensional, symmetric structures that support emergent personalization.

6.7 Implications for Organizational Design

The preceding results and discussion establish that organizational topology is a first-order determinant of personalization capability. This section translates that finding into practical guidance for organizational leaders, technology architects, and researchers studying complex adaptive organizations.

6.7.0.1 Topology-Aware Organizational Diagnostics

Most organizations already generate the raw data needed to compute their topological structure. Email headers, Slack channel membership, meeting co-attendance logs, Jira ticket handoffs, and pull-request review graphs each encode a communication network from which a simplicial complex can be constructed. We propose a *topological health check* consisting of three steps:

1. **Graph extraction.** Aggregate communication metadata over a rolling window (e.g. 90 days) to build a weighted, directed graph G_{org} . Vertices are teams or individuals; edges are weighted by message volume or co-event frequency.
2. **Complex construction and filtration.** Lift G_{org} to a clique complex and compute the sublevel filtration described in Section 3.2.
3. **Betti-number diagnostics.** Compute persistent homology and extract the summary statistics: $b_0^{(\infty)}$, TP_1 , TP_2 , k^* , and the platonic symmetry score $\mathcal{S}$ (Eq. (6)). Compare against the simulation benchmarks (Table 4).

This diagnostic is not a surveillance tool. It operates on aggregate structural patterns—who communicates with whom, and with what higher-order co-occurrence—not on message content. The output is a topological summary that reveals silos ($b_0 > 1$), approval loops ($b_1 > 0$), and strategy voids ($b_2 > 0$) at a structural level, enabling targeted intervention.

6.7.0.2 From Org Charts to Topological Engineering

Traditional organizational design focuses on optimizing nodes: hiring better talent, restructuring reporting lines, and redefining team mandates. These are important, but the simulation results suggest that the *space between nodes*—the morphisms, not the objects—deserves equal attention.

We propose the concept of *topological surgery*: the deliberate modification of an organization’s communication complex to fill pathological low-dimensional holes while preserving the high-dimensional coordination structure. Concrete actions include:

- **Instantiate higher-order simplices.** Create cross-functional pods (e.g. a weekly “personalization cell” comprising data science, engineering, merchandising, and UX) that constitute k -simplices for $k \geq 2$. These pods should have a shared deliverable, not merely a shared calendar invite.
- **Dissolve persistent 1-cycles.** Identify standing meetings or approval chains that form circular dependencies ($A \rightarrow B \rightarrow C \rightarrow A$) and replace them with direct integration events that collapse the loop into a higher-dimensional simplex.
- **Establish type-safe data contracts.** Implement cross-team interface agreements analogous to the type constraints of Section 2.2: define what data each team produces (writes) and consumes (reads), ensuring that hand-offs preserve semantic integrity without requiring every team to understand every other team’s internal language.

6.7.0.0.3 Incentive Alignment as Selection Pressure In the simulation, the reward function R_t served as a unified selection pressure that drove the topology toward its optimal configuration. In real organizations, the analog of R_t is the incentive structure.

When each functional team optimizes its own key performance indicator (e.g. click-through rate for merchandising, page-load time for engineering, brand sentiment for marketing), the organizational system is subject to competing selection pressures that pull the topology toward multiple, incompatible local optima. The result is a fragmented topology riddled with the low-dimensional holes identified by H3.

The simulation suggests that a necessary condition for topological self-organization is a shared, end-to-end metric that spans functional boundaries—for example, customer lifetime value, multi-touch journey completion rate, or long-horizon conversion. Such a metric aligns the selection pressure across the organization and allows the communication topology to evolve toward globally optimal structures.

6.8 Scalability Considerations

Platonic-symmetric topologies may have attractive scaling properties: many symmetric interconnection networks achieve $O(\log |V|)$ diameter [11], meaning that information traversal could grow logarithmically with organization size. A 1,024-person organization exhibiting such properties at the appropriate dimension might have a maximum communication path on the order of 10 hops—competitive with the flattest hierarchies. These scaling claims remain untested beyond the small simulated population used in this study.

However, the simulation used a relatively small agent population ($N = [\text{TBD}]$), and the scaling behavior of the platonic-convergence finding remains to be validated empirically. For very large organizations, a promising architectural pattern is the *hierarchically nested platonic approximation*: recursive composition of smaller platonic-symmetric units, where each node of a macro-level structure is itself a micro-level platonic-symmetric cluster. This retains the logarithmic-diameter property at each level of the hierarchy while remaining decoupled from any specific graph archi-

ture. We leave the formal analysis and simulation of hierarchical platonic approximations to future work.

7 Limitations and Future Work

7.1 Simulation Validity

The multi-agent simulation, while controlled and reproducible, involves several idealizations that limit direct generalization to real organizations.

- **Agents are not humans.** LLM-backed agents lack politics, ego, fatigue, institutional memory, and the informal relationships that shape real organizational communication. They comply perfectly with type constraints and respond deterministically to prompts—conditions that no human workforce fully satisfies.
- **Idealized type constraints.** Real organizations have messy, informal, and frequently violated communication norms. The clean interface contracts of Section 2.2 represent an upper bound on the degree of type safety achievable in practice.
- **Proxy reward.** The composite reward R_t is a proxy for customer lifetime value, not actual CLV. Its three components (task completion, response coherence, latency) capture important dimensions but do not exhaust the construct.
- **Synthetic consumer intent.** Intent episodes are sampled from a parametric mixture distribution, not drawn from real behavioral data. The distributional assumptions may not capture the full complexity of real consumer intent.

7.2 Topological Assumptions

- **Clique complex assumption.** The flag-complex construction assumes that all members of a clique interact simultaneously. In reality, higher-order interactions may be partial: three team members may share a Slack channel without all three reading every message. Weighted or probabilistic simplicial constructions [6] may better approximate real higher-order interactions.
- **Platonic symmetry score sensitivity.** The symmetry score $\mathcal{S}(\mathcal{K})$ (Eq. (6)) depends on the choice of kernel and reference complex. Future work should explore its sensitivity to these choices and validate it against alternative proximity metrics such as Gromov–Hausdorff distance to the nearest maximally symmetric complex.
- **Simplicial complex type as analytical strength.** We tested simplicial (clique) complexes; other types (directed flag complexes, path complexes) may capture different structural aspects of organizational communication. Importantly, simplicial complexes are the natural setting for platonic symmetry analysis, as they admit the full machinery of homology and kernel alignment—making this “limitation” also a structural advantage of the chosen representation.

7.3 Generalizability

- **Industry context.** The simulation models a retail e-commerce organization. Industries with different coordination requirements (e.g. healthcare, financial services,

manufacturing) may exhibit different optimal topologies or converge to different platonic attractors.

- **Role simplification.** Five functional roles is a deliberate simplification. Real organizations contain dozens of specialized teams whose interaction patterns may not decompose cleanly into five archetypes.
- **Cultural and geographic factors.** The simulation does not model cultural norms, time-zone constraints, or language barriers, all of which shape real communication topologies.

7.4 Future Research Directions

Empirical topological mapping. Apply TDA to internal communication logs (email metadata, Slack interaction graphs, meeting co-attendance matrices) from large enterprises and correlate the resulting Betti numbers and platonic symmetry scores with external performance indicators such as CLV, Net Promoter Score, and multi-touch conversion rates.

Homotopic transition costs. Develop a formal cost model for “topological surgery”: given an organization’s current communication complex $\mathcal{K}_{\text{now}}$ and a target topology exhibiting platonic symmetry, compute the minimum number of edge additions, removals, and higher-order coordination events required to deform one into the other, weighted by organizational disruption cost.

Dynamic topology and orchestrator agents. Extend the simulation to include real-time topology adaptation: “orchestrator agents” that observe incoming consumer-intent signals and re-route organizational workflows on the fly, matching the topology’s dimensionality to the signal’s complexity.

Theoretical bounds on platonic attractors. Pursue a formal characterization of the geometry of platonic attractors in simplicial complex space—which structures minimize a combined latency-expressivity objective among all complexes with a given number of vertices and edges? This extends the PRH programme from representation spaces to organizational topology and draws on results from distributed computing theory, where symmetric interconnection networks are well studied [11].

Scale experiments. Run the MAS simulation with 50–100+ agents across multiple role configurations to test whether the platonic-convergence finding persists at scale, and whether hierarchically nested platonic approximations emerge naturally in larger systems.

8 Conclusion

Digital personalization has been framed, for over a decade, as a prediction problem: build a better model, test it with a controlled experiment, and deploy the winner. This paper has argued that the prediction-centric view overlooks the organizational system through which personalized experiences are actually assembled and delivered. When the topology of information flow across functional teams is poorly structured, even optimal models cannot produce coherent, individualized consumer journeys.

We have made four contributions toward an alternative framing. First, we proposed a formal reframing of personalization as an emergent property of organizational information-flow topology (Section 2). Second, we constructed a multi-agent simulation in which role-specialized LLM agents coordinate under type-safe interface contracts, with the communication graph optimized via reinforcement-driven edge rewiring (Section 3.1). Third, we applied persistent homology to the evolving interaction networks, providing interpretable topological diagnostics—Betti numbers, persistence diagrams, and platonic symmetry scores—that characterize the organization’s structural capacity for integration (Section 3.2). Fourth, we reported an empirical finding that, under sustained selection pressure, organizational topologies converge toward a *latent platonic attractor*—characterized by a single connected component ($b_0 = 1$), no persistent lower-dimensional pathologies, and a single robust high-dimensional topological feature—consistent with the Platonic Representation Hypothesis [10] (Section 5.4).

The central insight is geometric. When the topology controller—given no architectural prior—is asked only to maximize personalization performance, it is predicted to converge on the platonic attractor: a structural limit consistent with the shared attractors toward which brains and language models converge under sufficient scale and task diversity [10]. Consistent with PRH: optimization under diverse constraints is expected to produce coordination patterns that were not explicitly designed, demonstrating emergence through adaptive self-organization under a unified selection pressure.

These findings suggest that the next frontier of competitive advantage in personalization lies not in better algorithms or richer data, but in the deliberate engineering—or, more precisely, the deliberate *cultivation*—of organizational topologies that support high-dimensional cross-functional integration. Personalization is not an algorithm. It is a topology.

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